Shawn N. Cummings, Ph.D.
Department of Language Science, University of California, Irvine
https://shawnncummings.github.io/ | scummin2@uci.edu | he/him/his

EDUCATION

Ph.D., Psychological Sciences (Language and Cognition) 2025
Joint affiliation with Speech, Language, and Hearing Sciences
University of Connecticut; Advisor: Dr. Rachel Theodore
Dissertation: Linking lexically guided perceptual learning to statistical patterns in speech input

M.S., Psychological Sciences (Language and Cognition) 2024
University of Connecticut; Advisor: Dr. Rachel Theodore
Thesis: Potential influences of autistic traits and perceptual acuity on lexically guided perceptual learning

Graduate Certificate in Neurobiology of Language 2024
University of Connecticut; Advisor: Dr. William Snyder

Graduate Certificate in Cognitive Science 2023
University of Connecticut; Advisor: Dr. Dimitris Xygalatis

B.A., Linguistics, University of Rochester 2021
Minors in Brain and Cognitive Sciences, English (Theatre), and Music

ACADEMIC EMPLOYMENT

Postdoctoral researcher, Language Science 2025
University of California, Irvine; PIs: Drs. Xin Xie and Chigusa Kurumada

HONORS AND AWARDS

DOE Graduate Assistance in Areas of National Need Fellowship 2023-2025
NSF NRT Traineeship: Science of Learning & Art of Communication 2022-2023
NIH T32 Fellowship: Cognitive Neuroscience of Communication 2021-2023
Donald Shankweiler Language Sciences Award (University of Connecticut) 2021

PEER REVIEWED ARTICLES AND PROCEEDINGS

* = equal contributions; + = undergraduate student I mentored

Cummings, S.N., Jaeger, T.F., Kurumada, C., & Xie, X. (2026). Adaptive Speech Perception: Empirical Indeterminacy and a Path Forward. *Proceedings of the Society for Computation in Linguistics.*

Cummings, S.N., Hodges, E.+, & Theodore, R. M. (2026). Cumulative input sensitivity predicts both attenuation and stability of lexically guided perceptual learning. *Psychonomic Bulletin & Review.* [OSF]

Cummings, S.N., Karboga, G., Yang, M., & Jaeger, T.F. (2025). Compensation in audiovisual speech perception: discounting the pen in the mouth. *Journal of Experimental Psychology*. [OSF]

Mechtenberg, H.*, **Cummings, S. N.***, Myers, E. B., & Luthra, S. (2025). Cents and shenshibility: The role of reward in talker-specific phonetic recalibration. *Attention, Perception, & Psychophysics*, 1-22. [OSF]

Cummings, S.N., Duda, B.+, & Theodore, R. M. (2025). Autistic listeners demonstrate robust lexically guided perceptual learning. *Autism Research.* [OSF]

Cummings, S. N., & Theodore, R. M. (2023). Hearing is believing: Lexically guided perceptual learning is graded to reflect the quantity of evidence in speech input. *Cognition*, *235*, 105404. [OSF]

Cummings, S. N., & Theodore, R. M. (2022). Perceptual learning of multiple talkers: Determinants, characteristics, and limitations. *Attention, Perception, & Psychophysics*, *84*(7), 2335-2359. [OSF]

ARTICLES UNDER REVIEW

Cummings, S.N., & Xie. X. (under review). Phonetic Boundary Calibration: A Rapid Method for Estimating Listener-Specific Phonetic Boundaries. [Gorilla open materials]

Basu, S., Famularo, L., **Cummings, S.N.**, Kurumada, C., & Xie, X. (under review). Toward Distributional Analyses of Accented Speech Production: The ADAPT Corpus and a Fricative Voicing Case Study. [OSF]

Cummings, S. N., Udell, A.+, & Theodore, R. M. (under review). Lexically guided perceptual learning is not a task-based artifact. [OSF]

Hodges, E.+, **Cummings, S.N.**, & Theodore, R. M. (In revision). Lexically guided perceptual learning is robust to linguistically engaging distraction. [OSF]

INVITED TALKS

Cummings, S.N. Modelling adaptive changes in speech perception. University of Rochester, September 2025.

Cummings, S.N. Perceptual learning and the statistics of speech input. Basque Center on Cognition, Brain, and Language, September 2025.

CONFERENCE PRESENTATIONS

Basu, S., Famularo, L., **Cummings, S.N.,** Kurumada, C., & Xie, X. (2027, January). Distributional Structure of Fricative Voicing in Mandarin-Accented English: Evidence from the ADAPT Corpus. Accepted for poster presentation at the International Workshop on Language Production, Hong Kong.

Cummings, S.N.*, Basu, S.*, Kurumada, C., & Xie, X. (2026, November). Toward Distributional Analyses of Accented Speech Production: The ADAPT Corpus and a Fricative Voicing Case Study. Accepted for talk at the 9th California Meeting on Psycholinguistics, Santa Barbara, California.

Ferrer, G.+, Lee, K.+, **Cummings, S.N.,** & Xie, X. (2026, November). How perceptual is lexically guided perceptual learning? Accepted for talk at the 9th California Meeting on Psycholinguistics, Santa Barbara, California.

Cummings, S.N., Jaeger, T.F., Kurumada, C., & Xie, X. (2026, July). Adaptive Speech Perception: Empirical Indeterminacy and a Path Forward. Poster presentation at the 9th meeting of the Society for Computation in Linguistics, San Diego, California.

Ferrer, G.+, Lee, K.+, **Cummings, S.N.,** & Xie, X. (2026, June). Probing the cognitive loci of compensation for co-articulation. Poster presentation at the 20th meeting of the Association for Laboratory Phonology, Montreal, Ontario, Canada.

Cummings, S.N., & Jaeger, T.F. (2026, June). No specialized mechanisms needed? Standard perceptual compensation might explain reduced perceptual learning. Poster presentation at the 20th meeting of the Association for Laboratory Phonology, Montreal, Ontario, Canada.

Cummings, S.N., & Xie, X. (2026, May). A novel paradigm for rapid determination of listener-specific phonetic boundaries. Poster presentation at the 190th Meeting of the Acoustical Society of America, Philadelphia, Pennsylvania.

Yuting Gu, **Cummings, S.N.**, Michelle Zhao, & Xie, X. (2026, May). Disentangling lexical decision criterion shifts from phonetic category changes in natural accent adaptation. Poster presentation at the 190th Meeting of the Acoustical Society of America, Philadelphia, Pennsylvania.

Jacob Bower, Yuting Gu, **Cummings, S.N.**, & Xie, X. (2026, May). Perception of Mandarin-Accented /s/-/θ/ Among L1-Mandarin Learners of English. Accepted for poster presentation at the 190th Meeting of the Acoustical Society of America, Philadelphia, Pennsylvania.

Cummings, S. N., & Theodore, R. M. (2024, November). Lexically guided perceptual learning is not a task-based artifact. Oral presentation at the 65th Annual Meeting of the Psychonomic Society, New York, New York.

Cummings, S. N. (2024, November). Lexically guided perceptual learning is robust! But what is it? Oral presentation at the EDULANG meeting on Variation and Diversity in Language Acquisition and Processing, Storrs, Connecticut.

Cummings, S. N., Duda, B.+, Medeiros, W., & Theodore, R. M. (2024, June). Potential influences of autistic traits and perceptual acuity on lexically guided perceptual learning. Oral presentation at the 9th Conference of the Scandinavian Association for Language and Cognition, Trondheim, Norway.

Cummings, S. N., Duda, B.+, Medeiros, W., & Theodore, R. M. (2024, May). Potential influences of autistic traits and perceptual acuity on lexically guided perceptual learning. Poster presentation at the 186th Meeting of the Acoustical Society of America, Ottawa, Ontario, Canada.

Giovannone, N., **Cummings, S. N.,** García-Sierra, A., Magnuson, J. S., & Theodore, R. M. (2023, November). Early neural encoding of acoustic-phonetic information is consistent across language ability. Poster presentation at the 64th Annual Meeting of the Psychonomic Society, San Francisco, CA.

Giovannone, N., **Cummings, S. N.,** García-Sierra, A., Magnuson, J. S., & Theodore, R. M. (2023, October). Early neural encoding of acoustic-phonetic information is consistent across language ability. Poster presentation at the 15th Annual Meeting of the Society for the Neurobiology of Language, Marseille, France.

Cummings, S. N., Choi, J. Y., Shattuck-Hufnagel, S., & Theodore, R. M. (2023, May). Linking lexically guided perceptual learning to statistical patterns in speech input. Poster presentation at the 184th Meeting of the Acoustical Society of America, Chicago, Illinois.

Hodges, E.+, **Cummings, S. N.,** & Theodore, R. M. (2023, May). Easy come, easy go: Examining the stability of lexically guided perceptual learning over time. Poster presentation at the 184th Meeting of the Acoustical Society of America, Chicago, Illinois.

Cummings, S.N., Karboga, G., Yang, M., & Jaeger, T.F. (2021, November) Causal Inference in the Perception and Learning of Audio-Visually Presented Speech. Poster presentation at the 62nd Annual Meeting of the Psychonomic Society.

TEACHING EXPERIENCE

* = course designed by me

Instructor:

*SLHS 6367: Topics in Hearing and Speech Science	Spring 2025
UNIV 1800: University Learning Skills	Fall 2024

Teaching Assistant:

SLHS 2156Q: Speech and Hearing Science	Fall 2023, 2024, Spring 2024, 2025 *Designated co-instructor F24 and S25*

Guest Lectures:

University of California, Irvine: Experimental Methods for Language Research	Fall 2026
University of Rochester: Speech Perception as an Adaptive Mechanism	Fall 2025, 2026
University of California, Irvine: Advanced Skills for Language Science III	Spring 2026
University of Connecticut: Seminar in Career Skills For Life After Grad School	Spring 2026
University of California, Irvine: Introduction to Speech Science	Fall 2025
University of Connecticut: Current Topics in Developmental Psychology	Spring 2024
University of Rochester: Undergraduate Research in Cognitive Science	Spring 2022

SERVICE

Mentor (3 students) 2024-2026
Mentoring Aspiring Graduate students and building an Inclusive Community (MAGIC)

Ad hoc Reviewer
Applied Psycholinguistics; Glossa Psycholinguistics; Journal of the Acoustical Society of America; Language and Speech; Language, Cognition, and Neuroscience; Language Learning; Journal of Experimental Psychology: General; Journal of Experimental Psychology: Human Perception and Performance; Attention, Perception, & Psychophysics

Reviewer
California Meeting on Psycholinguistics 2026
National Council on Undergraduate Research 2023

Student Facilitator 2023-2024
"Talk Shop" series, University of Connecticut

Poster Session Chair 2023
186th meeting of the Acoustical Society of America

Member, J-Term Committee 2021-2022
Science of Learning and Art of Communication Program, University of Connecticut

Member, Organizing Committee 2021
LangFest, University of Connecticut

SKILLS

Programming: R (proficient), Python (proficient), Stan (novice), Javascript (novice), MATLAB (novice)
Experimental design: Gorilla (proficient), OpenSesame (novice), PsychoPy (novice)
Acoustic analysis: Praat (proficient, annotation and scripting)
Electrophysiology recording: Brainvision product suite (proficient)
Language: English (native fluency), American Sign Language (intermediate), Spanish (intermediate)